2520030117

SKILL – 17

Active Job Listing and Foreground Switching

Aim

To implement active job listing, display status information, retrieve job IDs, format output, update state information, and test job listings.

Objective

- To list active jobs.
- To display job status information.
- To retrieve job IDs.
- To format job output.
- To update job state information.
- To test job listings.
- To identify the target job.
- To transfer terminal control.
- To resume processes.
- To wait for process completion.
- To update job state.
- To test foreground switching.

Problem Statement

To implement job-listing and foreground-switching functionality in a shell. The shell should identify active jobs, display their status and job IDs, transfer terminal control to a selected job, resume the process, wait for its completion, and update its state.

Theory

1. Active Job Listing
Displays the currently active jobs maintained by the shell.

2. Job ID
Each background job is assigned a job ID that can be used to identify and manage it.

3. Job Status
The shell maintains information about whether a job is running, stopped, or completed.
4. Foreground Switching
A selected background job can be brought to the foreground by transferring terminal control to it.

5. Process Completion
The shell waits for the foreground process to complete and then updates its job state.

Algorithm

1. Start the shell.
2. Create and maintain the job table.
3. Launch background jobs.
4. Store their job IDs and process information.
5. List the active jobs.
6. Display job status information.
7. Identify the target job.
8. Select the required job.
9. Transfer terminal control to the selected process.
10. Resume the process.
11. Wait for the process to complete.
12. Update the job state.
13. Display the updated job information.
14. Test foreground switching.

Data Used

Job IDs, process IDs, process groups, job states, commands, and terminal-control information are used.

Operations Used

- `fork()`
- `execvp()`
- `waitpid()`
- Process ID management
- Process group management
- Job table management
- Terminal control

Job Management Structure

Active Jobs
↓
Retrieve Job ID
↓
Identify Target Job
↓
Transfer Terminal Control
↓
Resume Process
↓

Wait for Completion

↓

Update Job State

Testing

- Test listing of active jobs.
- Verify job IDs.
- Verify job status information.
- Test selecting a target job.
- Test transferring a job to the foreground.
- Test process completion.
- Verify the updated job state.

Observation

- Active jobs are listed correctly.
- Job IDs are displayed.
- Job status information is maintained.
- The target job can be identified.
- Terminal control can be transferred to the foreground job.
- Job state is updated after completion.

Result

Active job listing and foreground switching were successfully implemented and tested.

Conclusion

The experiment demonstrates how a shell can list and manage active jobs, identify a target job, transfer terminal control to a foreground process, wait for completion, and update the job state.







